Optimal Placement of Extracellular Recording Electrodes Near Axon Tracts: a Note

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June 18, 2024

Abstract

In this note, the authors propose to consider the problem of optimal placement of electrodes when recording extracellularly from the vicinity of axons in an axon tract.

Some of the factors which are important to consider in placing recording electrodes include the way we model the extracellular field. Destexhe et. al. propose to model it as a low-pass filter [1]. One can ask where the tip should be placed for acquiring the best signal. Should it be placed between nodes or at nodes? In this context, the quadratic Gaussian CEO problem may be applicable [2].

In the forward problem the CEO is interested in L i.i.d. Gaussian data sequences which he cannot observe individually or directly. He can only observe a version of some combination of them. The electrode placement problem is somewhat dual to this setting. One can think of an electrode array with L electrodes, placed in the vicinity of the node which is the data source, and then apply a suitably modified version of the CEO problem analysis.

At the same time, this situation can be understood in greater detail by performing simulations with varying geometries as captured in W matrices and axonal inclinations, using the programs presented in [3]. The electrode array is placed in various configurations and the obtained signals are used to reconstruct the data source optimally. We hope to take this up in future work.

References

[1] Claude Bédard, Helmut Kröger, and Alain Destexhe. Modeling extracellular field potentials and the frequency-filtering properties of extracellular space. *Biophysical journal*, 86(3):1829–1842, 2004.

[2] Harish Viswanathan and Toby Berger. The quadratic gaussian ceo problem. *IEEE Transactions on Information Theory*, 43(5):1549–1559, 1997.

[3] Aman Chawla, Salvatore D. Morgera, and Arthur D. Snider. Fields, geometry, and their impact on axon interaction. *Journal of Applied Mathematics and Physics*, 9(4):751–778, 2021.